201514: metagranodiorite, Deep Well

(Big Bell Suite, Austin Downs Supersuite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Ninghan (SH 50-7), Maranalgo (2439)
MGA Zone 50, 563521E 6773105N

Sampled on 9 February 2017

This sample was collected from outcrop adjacent to the Paynes Find – Yalgoo Road on Oudabunna Station, about 5.2 km south of Bullamania Well, 5.1 km west of the Great Northern Highway, and 2.9 km north-northeast of Deep Well.

Geological context

The unit sampled is a monzogranite of the Big Bell Suite of the Austin Downs Supersuite (Van Kranendonk et al., 2013). The 2732–2686 Ma Big Bell Suite consists mainly of porphyritic and equigranular monzogranite and subordinate granodiorite and tonalite, together with their deformed and metamorphosed equivalents. The Big Bell Suite discordantly intruded mafic to felsic igneous and metasedimentary rocks of the Norie, Polelle and Glen Groups, as well as metagranitic and gabbroic rocks of the Annean Supersuite, and is intruded by granitic rocks of the Tuckanarra and Jungar Suites, and the Bald Rock Supersuite (Van Kranendonk et al., 2013; Zibra, 2019). A biotite metagranodiorite of the Big Bell Suite sampled about 79 km to the north yielded an igneous crystallization age of 2704 ± 5 Ma (GSWA 210258, Lu et al., 2018). A metamonzogranite collected about 1.0 km to the east-southeast yielded an igneous crystallization age of 2627 ± 3 Ma (GSWA 194754, Wingate et al., 2021).

Petrographic description

The sample is a medium-grained, equigranular metagranodiorite (Fig. 1), consisting of about 50–55% plagioclase, 25% quartz, 12% K-feldspar, 8% biotite, 2% magnetite and trace myrmekite and zircon. Plagioclase is anhedral to subhedral, equant to tabular, up to 2.5 mm in size, and weakly altered to sericite and clay minerals. Parts of the sample consist of aggregates of almost entirely equant plagioclase grains with a semi-polygonal texture. Quartz is anhedral, irregular, up to 4 mm in size, and partly encloses plagioclase. Quartz is weakly to moderately undulose but is not recrystallized. K-feldspar is anhedral, up to 8 mm in size, contains abundant inclusions of plagioclase and quartz, and displays microcline twinning. Biotite is forms subhedral laths up to 1 mm long, and shows minor replacement by chlorite, muscovite and trace epidote. Magnetite is subhedral, mainly equant, up to 0.8 mm in size, and is partially replaced by ilmenite. Although the sample lacks a tectonic foliation, the semi-polygonal texture, occurrence of microcline, and undulose extinction are consistent with solid-state deformation and recovery at low to moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are mainly colourless, subhedral to euhedral, and slightly rounded. The crystals are up to 250 µm long, and mostly elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and most crystals contain high-uranium, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 26–27 March 2021, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.22% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 18 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2710 ± 4 Ma (MSWD = 1.2).

The date of 2710 ± 4 Ma for the 18 analyses in Group I is interpreted as the igneous crystallization age of the granodiorite.

Figure 1. Hand specimen image of sample 201514: metagranodiorite, Deep Well

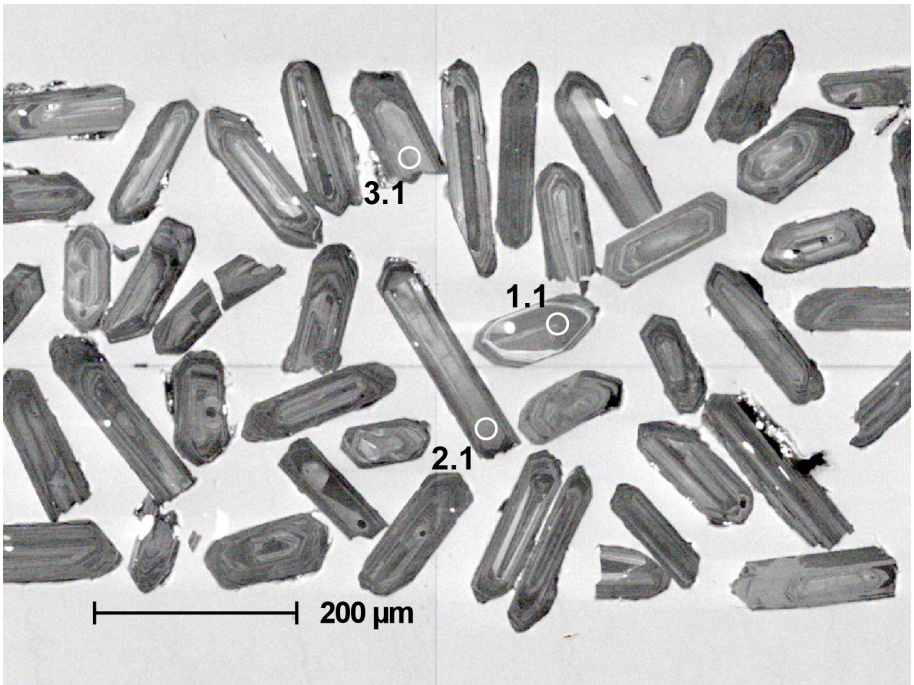

Figure 2. Cathodoluminescence image of representative zircons from sample 201514: metagranodiorite, Deep Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 201514: metagranodiorite, Deep Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	17	17.1	96	62	0.65	0.130	1.926	0.025	0.18515	0.00138	1.929	0.025	0.18399	0.00148	2693	29	2689	13	-0.1
I	9	9.1	199	211	1.06	0.419	2.003	0.020	0.18811	0.00097	2.011	0.020	0.18437	0.00120	2602	22	2693	11	3.4
I	3	3.1	164	96	0.59	0.050	1.988	0.022	0.18541	0.00112	1.989	0.022	0.18496	0.00115	2626	24	2698	10	2.7
I	15	15.1	133	81	0.61	0.438	2.046	0.023	0.18912	0.00123	2.055	0.023	0.18521	0.00150	2556	24	2700	13	5.3
I	7	7.1	800	78	0.10	0.010	1.953	0.015	0.18531	0.00052	1.954	0.015	0.18522	0.00052	2665	17	2700	5	1.3
I	6	6.1	91	63	0.69	0.029	1.886	0.025	0.18576	0.00147	1.886	0.025	0.18550	0.00150	2742	30	2703	13	-1.5
I	5	5.1	206	171	0.83	0.056	1.991	0.020	0.18628	0.00104	1.992	0.020	0.18578	0.00107	2623	22	2705	9	3.1
I	4	4.1	189	120	0.63	0.297	1.935	0.020	0.18863	0.00105	1.941	0.020	0.18598	0.00120	2679	23	2707	11	1.0
I	8	8.1	333	225	0.68	0.146	2.010	0.018	0.18730	0.00088	2.013	0.018	0.18600	0.00094	2600	20	2707	8	4.0
I	1	1.1	260	189	0.73	-0.010	1.940	0.018	0.18622	0.00089	1.940	0.018	0.18632	0.00089	2680	21	2710	8	1.1
I	10	10.1	166	114	0.69	-0.068	1.934	0.021	0.18586	0.00112	1.933	0.021	0.18646	0.00116	2688	24	2711	10	0.9
I	14	14.1	1234	537	0.44	0.025	1.997	0.014	0.18683	0.00046	1.997	0.014	0.18661	0.00047	2617	16	2713	4	3.5
I	11	11.1	116	73	0.63	0.023	1.994	0.024	0.18688	0.00130	1.995	0.024	0.18668	0.00132	2620	26	2713	12	3.4
I	18	18.1	87	32	0.37	0.446	1.882	0.026	0.19089	0.00152	1.890	0.026	0.18691	0.00184	2737	31	2715	16	-0.8
I	16	16.1	392	299	0.76	0.079	2.044	0.025	0.18766	0.00079	2.046	0.025	0.18695	0.00082	2565	26	2716	7	5.5
I	12	12.1	212	151	0.71	0.135	2.012	0.020	0.18822	0.00103	2.015	0.020	0.18702	0.00110	2598	22	2716	10	4.3
I	2	2.1	291	134	0.46	0.295	1.976	0.018	0.18988	0.00086	1.981	0.018	0.18725	0.00100	2634	20	2718	9	3.1
I	13	13.1	710	278	0.39	0.012	1.918	0.019	0.18773	0.00056	1.918	0.019	0.18763	0.00057	2705	22	2722	5	0.6

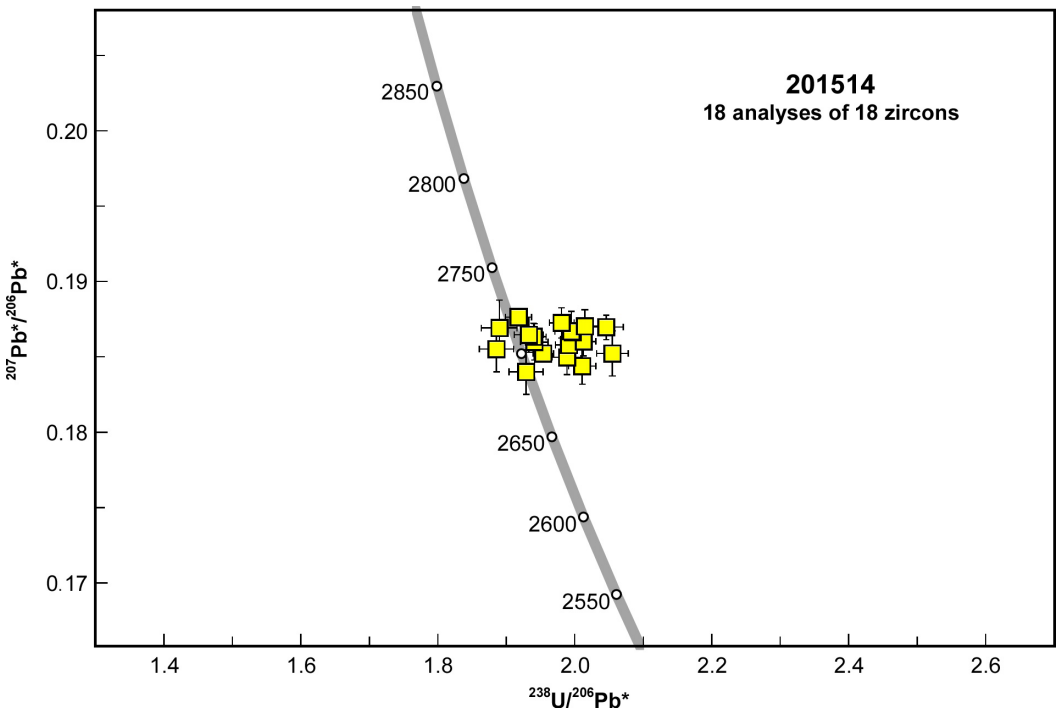

Figure 3. U–Pb analytical data for sample 201514: metagranodiorite, Deep Well. Yellow squares indicate Group I (magmatic zircons)

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Smithies, RH 2021, 201514: metagranodiorite, Deep Well; Geochronology Record 1821: Geological Survey of Western Australia, 4p.

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